

Architecture

Pharos RealFi Dashboard — Technical Architecture

Philosophy

The dashboard is built on three architectural principles:

1. **Zero dependency** — no npm, no webpack, no framework. A single HTML file runs the entire application.
 2. **Swap-ready data layer** — every piece of displayed data is isolated to a clearly marked location, ready to be replaced with a live RPC call in v1.1.
 3. **CSS-first performance** — animations, transitions, and visual effects are implemented in CSS wherever possible. JS is only used for data mutation (counter updates, feed rotation).
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File Structure

```
pharos-dashboard.html
|
|— <head>
|   └─ Google Fonts CDN link (DM Serif Display, Syne, JetBrains Mono)
|
|— <style> ─────────────────────────────────── ~350 lines
|   └─ :root { CSS custom properties }
|       └─ 10 semantic color tokens
|           └─ Referenced by every component
|
|   └─ body + ::before + ::after
|       └─ Background: 3-layer radial gradient + grid overlay
|
|   └─ nav, .logo, .network-badge, .wallet-btn
|   └─ .hero, .stats-row, .stat-card
|   └─ .main-grid, .left-col, .right-col
|   └─ .panel, .panel-header, .panel-title, .tab-row, .tab
```

```

|   ├── .chart-area, .chart-svg, .chart-labels
|   ├── .rwa-table, .asset-info, .asset-icon, .tag
|   ├── .staking-visual, .donut, .staking-stats, .stake-btn
|   ├── .network-grid, .net-card
|   ├── .dapp-grid, .dapp-card
|   ├── .feed, .feed-item, .feed-dot
|   ├── .realfi-banner, .btn-primary, .btn-secondary
|   ├── .tps-ticker, .tps-scroll
|   |
|   └─ @keyframes
|       pulse-glow      logo icon breathing glow
|       blink           network status dot
|       draw            SVG chart line stroke-dashoffset
|       donut-fill      staking donut stroke-dashoffset
|       fadeIn          feed item entrance
|       bar-grow        network pulse progress bars
|       ping            feed dot ripple
|       ticker          TPS marquee scroll
|
| └─ <body> ───────────────── semantic HTML
|   ├── <nav>             Logo · Badge · CTA
|   ├── .tps-ticker       Scrolling data marquee
|   ├── .hero             Title + 4 stat cards
|   ├── .main-grid
|   |   ├── .left-col
|   |   |   ├── Portfolio chart panel    (SVG inline)
|   |   |   ├── RWA Holdings panel        (<table>)
|   |   |   └─ dApp Explorer panel       (CSS grid cards)
|   |   └─ .right-col
|   |       ├── Staking panel             (SVG donut)
|   |       ├── Network Pulse panel       (metric cards)
|   |       └─ Activity Feed panel        (DOM list)
|   └─ .realfi-banner     CTA section
|
| └─ <script> ───────────────── ~55 lines
|   ├── Tab switcher      event listener, sibling deactivation
|   ├── TPS counter       setInterval(1800) → ±800 variance, clamp
|   ├── Portfolio drift   setInterval(3000) → ±$40 variance
|   └─ Feed rotator       setInterval(4000) → DOM prepend, max 6 items

```

CSS Architecture

Design Token System

```

:root {
  /* Backgrounds — darkest to lightest */
  --deep:      #040d18; /* body background */
  --ocean:     #071c30; /* deep panel */
  --mid:       #0a2540; /* mid-layer */
  --surface:   #0d2f4e; /* elevated surface */
  --glass:     rgba(13,47,78,0.55); /* glassmorphism panels */

  /* Accents */
  --accent:    #00d4ff; /* primary cyan — interactive, live indicators */
  --accent2:   #0af5c8; /* secondary teal — yields, positive values */
  --gold:      #f0b429; /* RWA gold — asset highlighting */
  --rose:      #ff6b6b; /* negative values, warnings */

  /* Text hierarchy */
  --muted:     #4a7a9b; /* labels, captions */
  --dim:       #6a9dbf; /* secondary text */
  --text:      #c8e6f5; /* body text */
  --white:     #f0f8ff; /* headings, emphasis */

  /* Structural */
  --border:    rgba(0,212,255,0.12); /* all borders */
  --glow:      0 0 30px rgba(0,212,255,0.15); /* panel glow */
}

```

Every color in the UI references one of these 10 tokens. Retheming the entire dashboard requires changing 10 values.

Glassmorphism Pattern

```

.panel {
  background: var(--glass); /* semi-transparent base */
  border: 1px solid var(--border); /* translucent cyan border */
  border-radius: 20px;
  backdrop-filter: blur(12px); /* background blur */
}

```

Animation Architecture

All entrance animations are triggered by page load. No scroll-triggered or interaction-triggered animations except hover states. This keeps the animation model simple and avoids IntersectionObserver complexity.

- └ SVG chart line draw (stroke-dashoffset, 2s, 0.3s delay)
- └ Staking donut fill (stroke-dashoffset, 1.6s)
- └ Network pulse bars (bar-grow, 1.5s, from width:0)
- └ Logo pulse-glow (infinite, 3s cycle)

Continuous (CSS) →

- └ TPS ticker scroll (ticker, 25s linear infinite)
- └ Network badge blink (blink, 2s infinite)
- └ Feed dot ping (ping, 2s infinite per dot)

Continuous (JS intervals) →

- └ TPS counter update (1800ms)
- └ Portfolio value drift (3000ms)
- └ Feed item rotation (4000ms)

SVG Chart Implementation

The portfolio chart is a fully custom inline SVG — no charting library.

```
<!-- Area fill -->
<path d="M0,140 C60,132 100,118 150,110 ..."
      fill="url(#chartGrad)"/>

<!-- Animated line (stroke-dashoffset trick) -->
<path d="M0,140 C60,132 100,118 150,110 ..."
      fill="none"
      stroke="url(#lineGrad)"
      stroke-width="2.5"
      filter="url(#glow)"
      stroke-dasharray="1200"
      stroke-dashoffset="1200"
      style="animation: draw 2s ease forwards 0.3s">
```

The `stroke-dasharray` equals the total path length. Setting `stroke-dashoffset` to the same value makes the line invisible. Animating it to 0 draws the line from left to right.

Data Layer (Mock → Live Upgrade Path)

Every data source in v1.0.0 is clearly isolated for future replacement.

TPS Counter (v1.0 mock → v1.1 live)

```

// CURRENT (v1.0 - mock)
let tps = 34218;
setInterval(() => {
  tps += Math.floor((Math.random() - 0.45) * 800);
  tps = Math.max(28000, Math.min(42000, tps));
  document.getElementById('live-tps').innerHTML = tps.toLocaleString() + ' <span>
}, 1800);

// FUTURE (v1.1 - live via Pharos RPC WebSocket)
const ws = new WebSocket('wss://rpc.pharos.xyz/ws');
ws.onmessage = (event) => {
  const data = JSON.parse(event.data);
  if (data.method === 'eth_subscription' && data.params.result.transactions) {
    const tps = data.params.result.transactions.length;
    document.getElementById('live-tps').innerHTML = tps.toLocaleString() + ' <spa
  }
};

```

Wallet Balance (v1.1 addition)

```

// Pharos Mainnet config
const PHAROS_RPC = 'https://rpc.pharos.xyz';
const CHAIN_ID = 688688;

async function connectWallet() {
  if (!window.ethereum) return alert('Install MetaMask or OKX Wallet');

  await window.ethereum.request({ method: 'eth_requestAccounts' });
  await window.ethereum.request({
    method: 'wallet_switchEthereumChain',
    params: [{ chainId: '0xA8008' }] // 688688 hex
  });

  const provider = new ethers.BrowserProvider(window.ethereum);
  const signer = await provider.getSigner();
  const address = await signer.getAddress();

  // Read PROS balance
  const prosBalance = await provider.getBalance(address);
  document.getElementById('port-val').textContent =
    '$' + (parseFloat(ethers.formatEther(prosBalance)) * PROS_PRICE).toFixed(0);
}

```

Responsive Design

Single breakpoint at 900px :

```
@media (max-width: 900px) {
  .stats-row { grid-template-columns: repeat(2, 1fr); } /* 4→2 cols */
  .main-grid { grid-template-columns: 1fr; }           /* 2→1 col */
  .hero-title { font-size: 30px; }                   /* scale down */
  .dapp-grid { grid-template-columns: 1fr; }          /* 2→1 col */
}
```

Below 900px, the right sidebar stacks below the left column. All panels remain functional.

Performance Budget

Metric	Value
HTML file size	~18KB
External requests	1 (Google Fonts)
JS execution at load	< 5ms
CSS animation count	10 keyframes
JS intervals running	3 (TPS, portfolio, feed)
DOM nodes	~180
Largest element	SVG chart (inline, ~1KB)

The dashboard scores **95+** on Lighthouse Performance when served over HTTP/2 with font preconnect.